



Utah State Standards Alignment

For Grades K-8

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About Zearn

Zearn is the 501(c)(3) nonprofit educational organization behind Zearn Math, the top-rated math learning platform used by 1 in 4 elementary-school students and by more than 1 million middle-school students nationwide. Zearn Math supports teachers with research-backed curriculum and digital lessons proven to double student academic growth across all levels of student proficiency. Zearn Math can be used flexibly to support a range of instructional needs including Tier 1 and 2 support, acceleration, intervention, tutoring, after school programming and summer programming.

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Zearn Math is aligned to Utah Core State Standards for Mathematics.

We are committed to ensuring that our high-quality instructional materials align to Utah Core State Standards for Mathematics to fully support Utah's students and teachers. The tables on the following pages, organized by Utah Core Standards, identify the alignment between each grade-level standard and Zearn Math lesson(s).

Table of Contents

Kindergarten	4
1st Grade	8
2nd Grade	12
3rd Grade	17
4th Grade	23
5th Grade	29
6th Grade	36
7th Grade	43
8th Grade	50

Kindergarten

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
Counting & Cardinality		
K.CC.1	Count to 100 by ones and by tens.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E
K.CC.2	Count forward beginning from a given number within the known sequence (instead of having to begin at 1).	Mission 1, Topic G Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E
K.CC.3	Read and write numbers using base ten numerals from 0 to 20. Represent a number of objects with a written numeral, in or out of sequence (0 represents a count of no objects).	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
K.CC.4	Understand the relationship between numbers and quantities; connect counting to cardinality. a. When counting objects, say the numbers in the standard order. Pair each quantity of objects with one and only one number, and each number with the correct quantity of objects. b. Understand that the last number said represents the number of objects counted. The number of objects is the same regardless of their arrangement or the order in which they were counted. c. Understand that each successive number refers to a quantity that is one greater than the previous number.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 1, Topic G Mission 1, Topic H Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
K.CC.5	Use counting to answer questions about “how many.” <i>For example, 20 or fewer objects arranged in a line, a rectangular array, or circle; 10 or fewer objects in a scattered configuration. Using a number from 1–20, count out that many objects.</i>	Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 5, Topic A

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
		Mission 5, Topic B Mission 5, Topic C Mission 5, Topic E
K.CC.6	Use matching or counting strategies to identify whether the number of objects in one group is greater than, less than, or equal to the number of objects in another group. Include groups with up to ten objects.	Mission 3, Topic B Mission 3, Topic E Mission 3, Topic F Mission 3, Topic G Mission 3, Topic H Mission 5, Topic E
K.CC.7	Compare two numbers between 1 and 10 presented as written numerals using “greater than,” “less than,” or “equal to.”	Mission 3, Topic F Mission 3, Topic G Mission 3, Topic H
Operations & Algebraic Thinking		
K.OA.1	Represent addition and subtraction with objects, fingers, mental images, simple drawings, or sounds. <i>For example, use clapping, act out situations, and use verbal explanations, expressions, or equations.</i>	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic G Mission 4, Topic H
K.OA.2	Solve addition and subtraction word problems within 10. Use objects or drawings to represent the problem.	Mission 4, Topic C Mission 4, Topic D Mission 4, Topic F Mission 4, Topic G Mission 4, Topic H
K.OA.3	Decompose numbers less than or equal to 10 into pairs in more than one way by using objects or drawings. Record each decomposition by a drawing or equation. <i>For example, $5 = 2 + 3$ and $5 = 4 + 1$.</i>	Mission 1, Topic C Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 4, Topic G
K.OA.4	Make sums of 10 using any number from 1 to 9. <i>For example, $2 + 8 = 10$.</i> Use objects or drawings to represent and record the answer.	Mission 4, Topic B Mission 4, Topic C Mission 4, Topic H
K.OA.5	Fluently add and subtract using numbers within 5.	Mission 4, Topic A

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
Numbers & Operations in Base Ten		
K.NBT.1	Compose and decompose numbers from 11–19 into ten ones and some further ones. Use objects or drawings and record each composition or decomposition by a drawing or equation. <i>For example, $18 = 10 + 8$. Understand that these numbers are composed of ten ones and one, two, three, four, five, six, seven, eight, or nine ones.</i>	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
Measurement & Data		
K.MD.1	Describe measurable attributes of objects, such as length or weight. Describe several measurable attributes of a single object.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic H
K.MD.2	Directly compare two objects with a measurable attribute in common, to see which object has "more of"/"less of" the attribute, and describe the difference. <i>For example, directly compare the length of two pencils and describe one as shorter or longer.</i>	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic F Mission 3, Topic F
K.MD.3	Classify objects into given categories; count the numbers of objects in each category and sort the categories by count. Limit the category counts to less than or equal to 10.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic E Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C
Geometry		
K.G.1	Describe objects in the environment using names of shapes, and describe the relative positions of these objects using terms such as above, below, beside, in front of, behind, and next to.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic B
K.G.2	Correctly name shapes regardless of their orientations or overall sizes.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
		Mission 6, Topic A
K.G.3	Identify shapes as two-dimensional ("flat") or three-dimensional ("solid").	Mission 2, Topic C
K.G.4	Analyze, compare, and sort two- and three-dimensional shapes and objects, in different sizes and orientations, using informal language to describe their similarities, differences, and other attributes (<i>for example, color, size, shape, number of sides</i>).	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic A Mission 6, Topic B
K.G.5	Model and create shapes from components such as sticks and clay balls.	Mission 6, Topic A
K.G.6	Compose simple shapes to form larger shapes. <i>For example, "Can you join these two triangles with full sides touching to make a rectangle?"</i>	Mission 6, Topic B

1st Grade

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
Operations & Algebraic Thinking		
1.OA.1	Use addition and subtraction within 20 to solve word problems involving situations of adding to, taking from, putting together, taking apart, and comparing, with unknowns in all positions. <i>For example, use objects, drawings, and equations with a symbol for the unknown number to represent the problem.</i>	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic G Mission 1, Topic H Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 3, Topic C Mission 3, Topic D Mission 4, Topic E Mission 6, Topic A Mission 6, Topic F
1.OA.2	Solve word problems that call for addition of three whole numbers whose sum is less than or equal to 20. <i>For example, use objects, drawings, and equations with a symbol for the unknown number to represent the problem.</i>	Mission 2, Topic A
1.OA.3	Apply properties of operations as strategies to add and subtract. <i>For example: If $8 + 3 = 11$ is known, then $3 + 8 = 11$ is also known. (Commutative property of addition.) To add $2 + 6 + 4$, the second two numbers can be added to make a ten, so $2 + 6 + 4 = 2 + 10 = 12$. (Associative property of addition.) First grade students need not use formal terms for these properties.</i>	Mission 1, Topic E Mission 1, Topic F Mission 2, Topic A Mission 2, Topic B
1.OA.4	Understand subtraction as an unknown-addend problem. <i>For example, subtract $10 - 8$ by finding the number that makes 10 when added to 8.</i>	Mission 1, Topic G Mission 1, Topic H Mission 1, Topic I Mission 2, Topic B Mission 2, Topic C
1.OA.5	Relate counting to addition and subtraction. <i>For example, by counting on 2 to add 2.</i>	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
		Mission 1, Topic D Mission 1, Topic G Mission 1, Topic H Mission 1, Topic I Mission 2, Topic B Mission 2, Topic C
1.OA.6	Add and subtract within 20. a. Use strategies such as counting on; making ten (<i>for example, $8 + 6 = 8 + 2 + 4 = 10 + 4 = 14$</i>); decomposing a number leading to a ten (<i>for example, $13 - 4 = 13 - 3 - 1 = 10 - 1 = 9$</i>); using the relationship between addition and subtraction (<i>for example, knowing that $8 + 4 = 12$, one knows $12 - 8 = 4$</i>); and creating equivalent but easier or known sums (<i>for example, adding $6 + 7$ by creating the known equivalent $6 + 6 + 1 = 12 + 1 = 13$</i>). b. By the end of Grade 1, demonstrate fluency for addition and subtraction within 10	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic F Mission 1, Topic I Mission 1, Topic J Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D
1.OA.7	Understand the meaning of the equal sign, and determine whether equations involving addition and subtraction are true or false. <i>For example, which of the following equations are true and which are false? $6 = 6$, $7 = 8 - 1$, $5 + 2 = 2 + 5$, $4 + 1 = 5 + 2$.</i>	Mission 1, Topic E Mission 2, Topic B Mission 2, Topic C
1.OA.8	Determine the unknown whole number in an addition or subtraction equation relating three whole numbers. <i>For example, determine the unknown number that makes the equation true in each of the equations $8 + ? = 11$, $5 = ? - 3$, $6 + 6 = ?$</i>	Mission 1, Topic D Mission 1, Topic H Mission 2, Topic C
Numbers & Operations in Base Ten		
1.NBT.1	Count to 120, starting at any number less than 120. In this range, read and write numerals and represent a number of objects with a written numeral.	Mission 4, Topic A Mission 6, Topic B
1.NBT.2	Understand that the two digits of a two-digit number represent amounts of tens and ones. Understand the following as special cases: a. 10 can be thought of as a bundle of ten ones, called a "ten." b. The numbers from 11 to 19 are composed of a ten and one, two, three, four, five, six, seven, eight, or nine ones. c. The numbers 10, 20, 30, 40, 50, 60, 70, 80, 90 refer to one,	Mission 2, Topic D Mission 4, Topic A Mission 4, Topic B Mission 4, Topic F Mission 6, Topic B

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
	two, three, four, five, six, seven, eight, or nine tens (and 0 ones).	
1.NBT.3	Compare two two-digit numbers based on meanings of the tens and ones digits, recording the results of comparisons with the symbols $>$, $=$, and $<$.	Mission 4, Topic B Mission 6, Topic B
1.NBT.4	Add within 100, including adding a two-digit number and a one-digit number, and adding a two-digit number and a multiple of 10, using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction; relate the strategy to a written method and explain the reasoning used. Understand that in adding two-digit numbers, one adds tens to tens and ones to ones, and that it is sometimes necessary to compose a ten.	Mission 4, Topic C Mission 4, Topic D Mission 4, Topic F Mission 6, Topic C Mission 6, Topic D
1.NBT.5	Given a two-digit number, mentally find 10 more or 10 less than the number, without having to count; explain the reasoning used.	Mission 2, Topic D Mission 4, Topic A Mission 6, Topic B
1.NBT.6	Subtract multiples of 10 in the range 10–90 from multiples of 10 in the range 10–90 (positive or zero differences), using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction; relate the strategy to a written method and explain the reasoning used.	Mission 4, Topic C Mission 6, Topic C
Measurement & Data		
1.MD.1	Order three objects by length; compare the lengths of two objects indirectly by using a third object.	Mission 3, Topic A Mission 3, Topic B
1.MD.2	Express the length of an object as a whole number of length units, by laying multiple copies of a shorter object (the length unit) end to end; understand that the length measurement of an object is the number of same-size length units that span it with no gaps or overlaps. Limit to contexts where the object being measured is spanned by a whole number of length units with no gaps or overlaps.	Mission 3, Topic B Mission 3, Topic C

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
1.MD.3	Tell and write time in hours and half-hours using analog and digital clocks.	Mission 5, Topic D
1.MD.4	Organize, represent, and interpret data with up to three categories; ask and answer questions about the total number of data points, how many in each category, and how many more or less are in one category than in another.	Mission 3, Topic D
1.MD.5	Identify the values of pennies, nickels, dimes and quarters, and know their comparative values. (<i>For example, a dime is of greater value than a nickel.</i>) Use appropriate notation to designate a coin's value. (<i>For example, 5¢.</i>)	Mission 4, Topic A Mission 6, Topic E
Geometry		
1.G.1	Distinguish between defining attributes (<i>for example, triangles are closed and three-sided</i>) versus non-defining attributes (<i>for example, color, orientation, overall size</i>); build and draw shapes that possess defining attributes.	Mission 5, Topic A
1.G.2	Compose shapes. a. Compose two-dimensional shapes (rectangles, squares, trapezoids, triangles, halfcircles, and quarter-circles) to create a composite shape, and compose new shapes from the composite shape. b. Compose three-dimensional shapes (cubes, right rectangular prisms, right circular cones, and right circular cylinders) to create a composite shape, and compose new shapes from the composite shape. First grade students do not need to learn formal names such as “right rectangular prism.”	Mission 5, Topic B
1.G.3	Partition circles and rectangles into two and four equal shares; describe the shares using the words halves, fourths, and quarters; and use the phrases half of, fourth of, and quarter of. Describe the whole as two or four of the shares. Understand that, for these examples, decomposing into more equal shares creates smaller shares.	Mission 5, Topic C Mission 5, Topic D

2nd Grade

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
Operations & Algebraic Thinking		
2.OA.1	Use addition and subtraction within 100 to solve one- and two-step word problems involving situations of adding to, taking from, putting together, taking apart, and comparing with unknowns in all positions, <i>for example, by using drawings and equations with a symbol for the unknown number to represent the problem.</i>	Mission 1, Topic B Mission 3, Topic G Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic F
2.OA.2	Fluently add and subtract within 20. a. Add and subtract within 20 using mental strategies such as counting on; making ten (<i>for example, $8 + 6 = 8 + 2 + 4 = 10 + 4 = 14$</i>); decomposing a number leading to a ten (<i>for example, $13 - 4 = 13 - 3 - 1 = 10 - 1 = 9$</i>); using the relationship between addition and subtraction (<i>for example, knowing that $8 + 4 = 12$, one knows $12 - 8 = 4$</i>); and creating equivalent but easier or known sums (<i>for example, adding $6 + 7$ by creating the known equivalent $6 + 6 + 1 = 12 + 1 = 13$</i>). b. By the end of Grade 2, know from memory all sums of two one-digit numbers.	Mission 1, Topic A Mission 1, Topic B
2.OA.3	Determine whether a group of objects (up to 20) has an odd or even number of members, (<i>for example, by pairing objects or counting them by twos</i>). Write an equation to express an even number as a sum of two equal addends.	Mission 6, Topic D
2.OA.4	Use addition to find the total number of objects arranged in rectangular arrays with up to 5 rows and up to 5 columns; write an equation to express the total as a sum of equal addends.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
Numbers & Operations in Base Ten		
2.NBT.1	Understand that the three digits of a three-digit number represent amounts of hundreds, tens, and ones; <i>for example, 706 equals 7 hundreds, 0 tens, and 6 ones</i> . Understand the following as special cases: a. 100 can be thought of as a bundle of ten tens called a "hundred." b. The numbers 100, 200, 300, 400, 500, 600, 700, 800, 900 refer to one, two, three, four, five, six, seven, eight, or nine hundreds (and 0 tens and 0 ones).	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
2.NBT.2	Count within 1,000; skip-count by fives, tens, and hundreds.	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 6, Topic A Mission 6, Topic B Mission 7, Topic B Mission 7, Topic E Mission 8, Topic D
2.NBT.3	Read and write numbers to 1,000 using base-ten numerals, number names, and expanded form.	Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
2.NBT.4	Compare two three-digit numbers based on meanings of the hundreds, tens, and ones digits, using $>$, $=$, and $<$ symbols to record the results of comparisons.	Mission 3, Topic E Mission 3, Topic F
2.NBT.5	Fluently add and subtract within 100 using strategies based on place value, properties of operations, and/or the relationship between addition and subtraction.	Mission 1, Topic B Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C
2.NBT.6	Add up to four two-digit numbers using strategies based on place value and properties of operations.	Mission 4, Topic D Mission 6, Topic A Mission 7, Topic B Mission 8, Topic D
2.NBT.7	Add and subtract within 1,000 using concrete models or drawings and strategies based on place value,	Mission 4, Topic B Mission 4, Topic C

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
	properties of operations, and/or the relationship between addition and subtraction; relate the strategy to a written method. Understand that in adding or subtracting three-digit numbers, one adds or subtracts hundreds and hundreds, tens and tens, and ones and ones, and that it is sometimes necessary to compose or decompose tens or hundreds.	Mission 4, Topic D Mission 4, Topic E Mission 4, Topic F Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D
2.NBT.8	Mentally add 10 or 100 to a given number 100-900, and mentally subtract 10 or 100 from a given number 100-900.	Mission 3, Topic G Mission 5, Topic A Mission 5, Topic D
2.NBT.9	Explain why addition and subtraction strategies work, using place value and the properties of operations. Explanations may be supported by drawings or objects.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 4, Topic F Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D
Measurement & Data		
2.MD.1	Measure the length of an object by selecting and using appropriate tools such as rulers, yardsticks, meter sticks, and measuring tapes.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 7, Topic C Mission 7, Topic D Mission 7, Topic F Mission 8, Topic A
2.MD.2	Measure the length of an object twice, using length units of different lengths for the two measurements; describe how the two measurements relate to the size of the unit chosen.	Mission 2, Topic C Mission 7, Topic D
2.MD.3	Estimate lengths using units of inches, feet, centimeters, and meters.	Mission 2, Topic B Mission 2, Topic D Mission 7, Topic D

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
2.MD.4	Measure to determine how much longer one object is than another, expressing the length difference in terms of a standard length unit. <i>For example, after measuring a pencil and a crayon, a student uses the measurements to determine that the pencil is two inches longer than the crayon.</i>	Mission 2, Topic C Mission 2, Topic D Mission 7, Topic D
2.MD.5	Use addition and subtraction within 100 to solve word problems involving lengths that are given in the same units. <i>For example, use drawings (such as drawings of rulers) and equations with a symbol for the unknown number to represent the problem.</i>	Mission 2, Topic D Mission 7, Topic E Mission 7, Topic F
2.MD.6	Represent whole numbers as lengths from 0 on a number line diagram with equally spaced points corresponding to the numbers 0, 1, 2... Represent whole number sums and differences within 100 on a number line diagram.	Mission 7, Topic A Mission 7, Topic E Mission 7, Topic F
2.MD.7	Tell and write time from analog and digital clocks to the nearest five minutes, using a.m. and p.m.	Mission 8, Topic D
2.MD.8	Solve word problems involving dollar bills, quarters, dimes, nickels, and pennies, using \$ and ¢ symbols appropriately. <i>For example, if you have 2 dimes and 3 pennies, how many cents do you have?</i>	Mission 3, Topic D Mission 7, Topic B
2.MD.9	Generate measurement data by measuring lengths of several objects to the nearest whole unit, or by making repeated measurements of the same object. Show the measurements by making a line plot, where the horizontal scale is marked off in whole-number units.	Mission 7, Topic F
2.MD.10	Draw a picture graph and a bar graph (with single-unit scale) to represent a data set with up to four categories. Solve simple put-together, take-apart, and comparison problems using information presented in a bar graph.	Mission 7, Topic A
Geometry		
2.G.1	Recognize and draw shapes having specified attributes, such as a given number of angles or a given	Mission 8, Topic A Mission 8, Topic B

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
	number of equal faces. Sizes are compared directly or visually, not compared by measuring. Identify triangles, quadrilaterals, pentagons, hexagons, and cubes.	Mission 8, Topic C
2.G.2	Partition a rectangle into rows and columns of same-size squares and count to find the total number of squares.	Mission 8, Topic C
2.G.3	Partition circles and rectangles into two, three, or four equal shares; describe the shares using the words halves, thirds, half of, a third of, etc.; and describe the whole as two halves, three thirds, or four fourths. Recognize that equal shares of identical wholes need not have the same shape.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D

3rd Grade

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
Operations & Algebraic Thinking		
3.OA.1	Interpret the products of whole numbers, such as interpreting 5×7 as the total number of objects in 5 groups of 7 objects each. <i>For example, describe a context in which a total number of objects can be expressed as 5×7.</i>	Mission 1, Topic A Mission 1, Topic C Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic F
3.OA.2	Interpret whole-number quotients of whole numbers. <i>For example, interpret $56 \div 8$ as the number of objects in each share when 56 objects are partitioned equally into eight shares (partitive), or as a number of shares when 56 objects are partitioned into equal shares of eight objects each (quotative).</i>	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.OA.3	Use multiplication and division within 100 to solve word problems in situations involving equal groups, arrays, and measurement quantities. <i>For example, use drawings and equations with a symbol for the unknown number to represent the problem.</i>	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.OA.4	Determine the unknown whole number in a multiplication or division equation relating three whole numbers. <i>For example, determine the unknown number— product, factor, quotient, dividend, or</i>	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
	divisor—that makes the equation true in each of the equations $8 \times ? = 48$, $5 = ? \div 3$, and $6 \times 6 = ?$	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.OA.5	Apply properties of operations as strategies to multiply and divide. <i>For example: If $6 \times 4 = 24$ is known, then $4 \times 6 = 24$ is also known (commutative property of multiplication). $3 \times 5 \times 2$ can be found by $3 \times 5 = 15$, then $15 \times 2 = 30$, or by $5 \times 2 = 10$, then $3 \times 10 = 30$ (associative property of multiplication). Knowing that $8 \times 5 = 40$ and $8 \times 2 = 16$, one can find 8×7 as $8 \times (5 + 2) = (8 \times 5) + (8 \times 2) = 40 + 16 = 56$ (distributive property). (Third grade students may, but need not, use formal terms for these properties.)</i>	Mission 1, Topic C Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic F
3.OA.6	Understand division as an unknown-factor problem. Understand the relationship between multiplication and division (multiplication and division are inverse operations). <i>For example, find $32 \div 8$ by finding the number that makes 32 when multiplied by 8.</i>	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.OA.7	Fluently multiply and divide. a. Fluently multiply and divide within 100, using strategies such as the relationship between multiplication and division or properties of operations. <i>(For example, knowing that $8 \times 5 = 40$, one knows $40 \div 5 = 8$.)</i> b. <i>By the end of Grade 3, know from memory all products of two one-digit numbers.</i>	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.OA.8	Solve two-step word problems. a. Solve two-step word problems using the four operations. Know how to perform operations in the conventional order when there are no parentheses to specify a particular order (Order of Operations). (Limit to problems posed with whole numbers and having whole number answers.) b. Represent two-step problems using equations with a letter standing for the unknown quantity. Create	Mission 1, Topic F Mission 3, Topic C Mission 3, Topic E Mission 3, Topic F Mission 7, Topic A

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
	accurate equations to match word problems. c. Assess the reasonableness of answers using mental computation and estimation strategies, including rounding.	
3.OA.9	Identify arithmetic patterns (including patterns in the addition table or multiplication table), and explain them using properties of operations. <i>For example, observe that four times a number is always even, and explain why four times a number can be decomposed into two equal addends.</i>	Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic F
Numbers & Operations in Base Ten		
3.NBT.1	Use place value understanding to round whole numbers to the nearest 10 or 100.	Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.NBT.2	Fluently add and subtract within 1,000 using strategies and algorithms based on place value, properties of operations, and/or the relationship between addition and subtraction.	Mission 2, Topic A Mission 2, Topic D Mission 2, Topic E
3.NBT.3	Multiply one-digit whole numbers by multiples of 10 in the range 10–90 (<i>for example, 9×80 and 5×60</i>) using strategies based on place value and properties of operations.	Mission 3, Topic F
Numbers & Operations - Fractions		
3.NF.1	Understand that a unit fraction has a numerator of one and a non-zero denominator. a. Understand a fraction $\frac{1}{b}$ as the quantity formed by one part when a whole is partitioned into b equal parts. b. Understand a fraction $\frac{a}{b}$ as the quantity formed by a parts of size $\frac{1}{b}$. <i>For example: $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$.</i>	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C
3.NF.2	Understand a fraction as a number on the number line; represent fractions on a number line diagram. a. Represent a fraction $\frac{1}{b}$ on a number line diagram by defining the interval from 0 to 1 as the whole and partitioning it into b equal parts. Recognize that each part has size $\frac{1}{b}$ and that the endpoint of the part	Mission 5, Topic D

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
	based at 0 locates the number $1/b$ on the number line. b. Represent a fraction a/b on a number line diagram by marking off a lengths $1/b$ from 0. Recognize that the resulting interval has size a/b and that its endpoint locates the number a/b on the number line.	
3.NF.3	Explain equivalence of fractions in special cases, and compare fractions by reasoning about their size. a. Understand two fractions as equivalent if they are the same size, or the same point on a number line. b. Recognize and generate simple equivalent fractions, such as $1/2 = 2/4$, $4/6 = 2/3$. <i>Explain why the fractions are equivalent by using a visual fraction model, for example.</i> c. Express whole numbers as fractions, and recognize fractions that are equivalent to whole numbers. <i>For example, express 3 in the form $3 = 3/1$; recognize that $6/1 = 6$; locate $4/4$ and 1 at the same point of a number line diagram.</i> d. Compare two fractions with the same numerator or the same denominator by reasoning about their size. Recognize that comparisons are valid only when the two fractions refer to the same whole. <i>Record the results of comparisons with the symbols $>$, $=$, or $<$, and justify the conclusions, for example, by using a visual fraction model.</i>	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
Measurement & Data		
3.MD.1	Tell and write time to the nearest minute and measure time intervals in minutes. Solve word problems involving addition and subtraction of time intervals in minutes, <i>for example, by representing the problem on a number line diagram.</i>	Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.MD.2	Measure and estimate liquid volumes and masses of objects using standard units of grams (g), kilograms (kg), milliliters (ml), and liters (l). (Excludes compound UTAH CORE STATE STANDARDS for MATHEMATICS GRADE 3 26 units such as cubic centimeters [cc or cm^3] and finding the geometric volume of a container.) <i>Add, subtract, multiply, or divide to solve one-step word problems involving masses of objects or volumes of liquids that are given in the same units, for example, by using drawings (such as a beaker with a</i>	Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
	<i>measurement scale) to represent the problem. (Excludes multiplicative comparison problems.)</i>	
3.MD.3	Draw a scaled picture graph and a scaled bar graph to represent a data set with several categories. Solve one- and two-step "how many more" and "how many less" problems using information presented in scaled bar graphs. <i>For example, draw a bar graph in which each square in the bar graph might represent five pets.</i>	Mission 6, Topic A
3.MD.4	Generate measurement data by measuring lengths using rulers marked with halves and fourths of an inch. Show the data by making a line plot where the horizontal scale is marked off in appropriate units-whole numbers, halves, or quarters.	Mission 6, Topic B Mission 7, Topic D
3.MD.5	Recognize area as an attribute of plane figures and understand concepts of area measurement. a. A square with side length one unit, called "a unit square," is said to have "one square unit" of area, and can be used to measure area. b. A plane figure which can be covered without gaps or overlaps by n unit squares is said to have an area of n square units.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
3.MD.6	Measure area by counting unit squares (square centimeters, square meters, square inches, square feet, and improvised units).	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic D
3.MD.7	Relate area to the operations of multiplication and addition (refer to 3.OA.5). a. Find the area of a rectangle with whole-number side lengths by tiling it, and show that the area is the same as would be found by multiplying the side lengths. b. Multiply side lengths to find areas of rectangles with whole-number side lengths in the context of solving real-world and mathematical problems, and represent whole number products as rectangular areas in mathematical reasoning. c. Use tiling to show in a concrete case that the area of a rectangle with whole-number side lengths a and $b + c$ is the sum of $a \times b$ and $a \times c$. Use area models to represent the distributive property in mathematical reasoning. d. Recognize area as additive. Find areas of rectilinear figures by decomposing them into non-overlapping rectangles	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
	and adding the areas of the non-overlapping parts, applying this technique to solve real-world problems.	
3.MD.8	Solve real-world and mathematical problems involving perimeters of polygons, including finding the perimeter given the side lengths, finding an unknown side length, and exhibiting rectangles with the same perimeter and different areas or with the same area and different perimeters.	Mission 7, Topic C Mission 7, Topic D Mission 7, Topic E
Geometry		
3.G.1	Understand that shapes in different categories (<i>for example, rhombuses, rectangles, and others</i>) may share attributes (<i>for example, having four sides</i>), and that the shared attributes can define a larger category (<i>for example, quadrilaterals</i>). Recognize rhombuses, rectangles, and squares as examples of quadrilaterals, and draw examples of quadrilaterals that do not belong to any of these subcategories.	Mission 7, Topic B
3.G.2	Partition shapes into parts with equal areas. Express the area of each part as a unit fraction of the whole. <i>For example, partition a shape into four parts with equal area, and describe the area of each part as 1/4 of the area of the shape.</i>	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C

4th Grade

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
Operations & Algebraic Thinking		
4.OA.1	Interpret a multiplication equation as a comparison (<i>for example, interpret $35 = 5 \times 7$ as a statement that 35 is 5 times as many as 7 and 7 times as many as 5</i>). Represent verbal statements of multiplicative comparisons as multiplication equations.	Mission 1, Topic A Mission 3, Topic A Mission 3, Topic B Mission 3, Topic D Mission 7, Topic A
4.OA.2	Multiply or divide to solve word problems involving multiplicative comparison, <i>for example, by using drawings and equations with a symbol for the unknown number to represent the problem, distinguishing multiplicative comparison from additive comparison</i> .	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 5, Topic G Mission 7, Topic A Mission 7, Topic B
4.OA.3	Solve multi-step word problems posed with whole numbers and having whole-number answers using the four operations, including problems in which remainders must be interpreted. a. Represent these problems using equations with a letter standing for the unknown quantity. b. Assess the reasonableness of answers using mental computation and estimation strategies, including rounding.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 3, Topic H Mission 7, Topic B Mission 7, Topic C
4.OA.4	Find all factor pairs for a whole number in the range 1-100. Recognize that a whole number is a multiple of each of its factors. Determine whether a given whole number in the range 1-100 is a multiple of a given one-digit number. Determine whether a given whole number in the range 1-100 is prime or composite.	Mission 3, Topic F
4.OA.5	Generate a number or shape pattern that follows a given rule. Identify apparent features of the pattern that were not explicit in the rule itself. <i>For example, given the rule "add 3" and the starting number 1, generate terms in the resulting sequence and observe that the terms appear to alternate between odd and</i>	Mission 5, Topic H

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
even numbers. Explain informally why the numbers will continue to alternate in this way.		
Numbers & Operations in Base Ten		
4.NBT.1	Recognize that in a multi-digit whole number, a digit in one place represents ten times what it represents in the place to its right. <i>For example, recognize that $700 \div 70 = 10$ by applying concepts of place value and division.</i>	Mission 1, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic G
4.NBT.2	Read and write multi-digit whole numbers using base-ten numerals, number names, and expanded form. Compare two multi-digit numbers based on meanings of the digits in each place, using $>$, $=$, and $<$ symbols to record the results of comparisons.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
4.NBT.3	Use place value understanding to round multi-digit whole numbers to any place.	Mission 1, Topic C
4.NBT.4	Fluently add and subtract multi-digit whole numbers using the standard algorithm.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
4.NBT.5	Multiply a whole number of up to four digits by a one-digit whole number, and multiply two two-digit numbers, using strategies based on place value and the properties of operations. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic H
4.NBT.6	Find whole-number quotients and remainders with up to four-digit dividends and one-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 3, Topic E Mission 3, Topic F Mission 3, Topic G
Numbers & Operations - Fractions		
4.NF.1	Explain why a fraction a/b is equivalent to a fraction $(n \times a)/(n \times b)$ by using visual fraction models, with	Mission 5, Topic B Mission 5, Topic D

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
	attention to how the number and size of the parts differ even though the two fractions themselves are the same size. Use this principle to recognize and generate equivalent fractions.	Mission 5, Topic E Mission 6, Topic B
4.NF.2	Compare two fractions with different numerators and different denominators, <i>for example, by creating common denominators or numerators, or by comparing to a benchmark fraction such as $\frac{1}{2}$</i> . Recognize that comparisons are valid only when the two fractions refer to the same whole. Record the results of comparisons with symbols $>$, $=$, or $<$, and justify the conclusions, <i>for example, by using a visual fraction model</i> .	Mission 5, Topic C Mission 5, Topic E
4.NF.3	Understand a fraction $\frac{a}{b}$ with $a > 1$ as a sum of fractions $\frac{1}{b}$. In other words, any fraction is a sum of unit fractions. a. Understand addition and subtraction of fractions as joining and separating parts referring to the same whole. b. Decompose a fraction into a sum of fractions with the same denominator in more than one way, recording each decomposition by an equation. Justify decompositions, <i>for example, by using a visual fraction model. For example, $\frac{3}{8} = \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$; $\frac{3}{8} = \frac{1}{8} + \frac{2}{8}$; $2\frac{1}{8} = 1 + 1 + \frac{1}{8}$; $2\frac{1}{8} = \frac{8}{8} + \frac{8}{8} + \frac{1}{8}$</i> . c. Add and subtract mixed numbers with like denominators, <i>for example, by replacing each mixed number with an equivalent fraction, and/or by using properties of operations and the relationship between addition and subtraction. For example, $3\frac{1}{4} + 2\frac{1}{4} = \frac{13}{4} + \frac{9}{4} = \frac{22}{4}$; $3\frac{1}{4} + 2\frac{1}{4} = (3 + 2) + (\frac{1}{4} + \frac{1}{4}) = 5 + \frac{2}{4} = 5\frac{2}{4}$, which is equivalent to $\frac{22}{4}$</i> . d. Solve word problems involving addition and subtraction of fractions referring to the same whole and having like denominators, <i>for example, by using visual fraction models and equations to represent the problem</i> .	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F Mission 6, Topic D
4.NF.4	Apply and extend previous understandings of multiplication to multiply a fraction by a whole number. a. Understand a fraction $\frac{a}{b}$ as a multiple of $\frac{1}{b}$. <i>For example, use a visual fraction model to represent $\frac{5}{4}$ as the product $5 \times (\frac{1}{4})$, recording the conclusion by the equation $\frac{5}{4} = 5 \times (\frac{1}{4})$</i> . b. Understand a multiple of $\frac{a}{b}$ as a multiple of $\frac{1}{b}$, and	Mission 5, Topic A Mission 5, Topic E Mission 5, Topic G

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
	use this understanding to multiply a fraction by a whole number. <i>For example, use a visual fraction model to express $3 \times (2/5)$ as $6 \times (1/5)$, recognizing this product as $6/5$. (In general, $n \times (a/b) = (n \times a)/b$).</i> c. Solve word problems involving multiplication of a fraction by a whole number (for example, by using visual fraction models and equations to represent the problem). <i>For example, if each person at a party will eat $3/8$ of a pound of roast beef, and there will be five people at the party, how many pounds of roast beef will be needed? Between what two whole numbers does your answer lie?</i>	
4.NF.5	Express a fraction with denominator 10 as an equivalent fraction with denominator 100, and use this technique to add two fractions with respective denominators 10 and 100. <i>For example, express $3/10$ as $30/100$, and add $3/10 + 4/100 = 34/100$.</i>	Mission 6, Topic B Mission 6, Topic D Mission 6, Topic E
4.NF.6	Use decimal notation for fractions with denominators 10 or 100. <i>For example, rewrite 0.62 as $62/100$, describe a length as 0.62 meters; locate 0.62 on a number line diagram.</i>	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic D Mission 6, Topic E
4.NF.7	Compare two decimals to hundredths by reasoning about their size. Recognize that comparisons are valid only when the two decimals refer to the same whole. Record the results of comparisons with the symbols $>$, $=$, or $<$, and justify the conclusions, <i>for example, by using a visual model.</i>	Mission 6, Topic B Mission 6, Topic C
Measurement & Data		
4.MD.1	Know relative sizes of measurement units within each system of units (standard and metric), including kilometers, meters, and centimeters; liters and milliliters; kilograms and grams; pounds and ounces; hours, minutes, and seconds. Within a single system of measurement, express measurements in a larger unit in terms of a smaller unit. Record measurement equivalents in a two-column table. <i>For example, know that one foot is 12 times as long as one inch. Express the length of a four-foot snake as 48 inches. Know that one meter is 100 times as long as one centimeter.</i>	Mission 2, Topic A Mission 2, Topic B Mission 5, Topic D Mission 5, Topic F Mission 5, Topic G Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 6, Topic D Mission 7, Topic A Mission 7, Topic B

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
	Generate a conversion table for feet and inches listing the number pairs (1, 12), (2, 24), (3, 36)...	Mission 7, Topic C
4.MD.2	Use the four operations to solve word problems involving distances, intervals of time, liquid volumes, masses of objects, and money. a. Include problems involving simple fractions or decimals, and problems that require expressing measurements given in a larger unit in terms of a smaller unit. b. Represent measurement quantities using diagrams such as number line diagrams that feature a measurement scale.	Mission 2, Topic A Mission 2, Topic B Mission 5, Topic D Mission 5, Topic F Mission 5, Topic G Mission 6, Topic C Mission 6, Topic E Mission 7, Topic A Mission 7, Topic B Mission 7, Topic C
4.MD.3	Apply the area and perimeter formulas for rectangles in real-world and mathematical problems. <i>For example, find the width of a rectangular room given the area of the flooring and the length, by viewing the area formula as a multiplication equation with an unknown factor.</i>	Mission 3, Topic A Mission 3, Topic H
4.MD.4	Make a line plot to display a data set of measurements in fractions of a unit (halves, quarters, and eighths). Solve problems involving addition and subtraction with like denominators of fractions by using information presented in line plots. <i>For example, use a line plot to find and interpret the difference in length between the longest and shortest pencils in a classroom.</i>	Mission 5, Topic E Mission 5, Topic G
4.MD.5	Recognize angles as geometric figures that are formed wherever two rays share a common endpoint, and understand concepts of angle measurement. a. Understand that an angle is measured with reference to a circle with its center at the common endpoint of the rays, by considering the fraction of the circular arc between the points where the two rays intersect the circle. An angle that turns through $\frac{1}{360}$ of a circle is called a "one-degree angle," and can be used to measure other angles. b. Understand that an angle that turns through n one-degree angles is said to have an angle measure of n degrees.	Mission 4, Topic B

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.MD.6	Measure angles in whole-number degrees using a protractor. Sketch angles of specified measure.	Mission 4, Topic B
4.MD.7	Recognize angle measure as additive. a. Understand that when an angle is decomposed into non-overlapping parts, the angle measure of the whole is the sum of the angle measures of the parts. b. Solve addition and subtraction problems to find unknown angles on a diagram in real-world and mathematical problems, <i>for example by using an equation with a symbol for the unknown angle measure.</i>	Mission 4, Topic C
Geometry		
4.G.1	Draw points, lines, line segments, rays, angles (right, acute, and obtuse), and perpendicular and parallel lines. Identify these in two-dimensional figures.	Mission 4, Topic A Mission 4, Topic D
4.G.2	Classify two-dimensional figures based on the presence or absence of parallel or perpendicular lines, or the presence or absence of angles of a specified size. Recognize right triangles as a category, and identify right triangles.	Mission 4, Topic D
4.G.3	Recognize a line of symmetry for a two-dimensional figure as a line across the figure such that the figure can be folded along the line into matching parts. Identify line-symmetric figures and draw lines of symmetry	Mission 4, Topic D

5th Grade

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
Operations & Algebraic Thinking		
5.OA.1	Use parentheses, brackets, or braces in numerical expressions, and evaluate expressions with these symbols.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 4, Topic D Mission 4, Topic G Mission 4, Topic H
5.OA.2	Write and interpret simple numerical expressions. a. Write simple expressions that record calculations with numbers. <i>For example, use $2 \times (8+7)$ to express the calculation "add 8 and 7, then multiply by 2."</i> b. Interpret numerical expressions without evaluating them. <i>For example, use conceptual understanding of multiplication to interpret $3 \times (18939 + 921)$ as being three times as large as $18939 + 921$ without calculating the indicated sum or product.</i>	Mission 2, Topic B Mission 2, Topic C Mission 4, Topic D Mission 4, Topic H Mission 6, Topic B
5.OA.3	Generate two numerical patterns using two given rules. Identify apparent relationships between corresponding terms. Form ordered pairs consisting of corresponding terms from the two patterns, and graph the ordered pairs on a coordinate plane. <i>For example, given the rule "add 3" and the starting number 0, and given the rule "add 6" and the starting number 0, generate terms in the resulting sequences, and observe that the terms in one sequence are twice the corresponding terms in the other sequence. Explain informally why this is so.</i>	Mission 6, Topic B Mission 6, Topic D
Numbers & Operations in Base Ten		
5.NBT.1	Recognize that in a multi-digit number, a digit in one place represents 10 times as much as it represents in the place to its right and 1/10 of what it represents in the place to its left.	Mission 1, Topic A Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
5.NBT.2	Explain patterns in the number of zeros of the product when multiplying a number by powers of 10, and explain patterns in the placement of the decimal point when a decimal is multiplied or divided by a power of 10. Use whole-number exponents to denote powers of 10.	Mission 1, Topic A Mission 1, Topic D Mission 1, Topic E Mission 2, Topic A Mission 2, Topic D Mission 2, Topic E Mission 2, Topic G
5.NBT.3	Read, write, and compare decimals to thousandths. a. Read and write decimals to thousandths using base-ten numerals, number names, and expanded form. <i>For example, $347.392 = 3 \times 100 + 4 \times 10 + 7 \times 1 + 3 \times (1/10) + 9 \times (1/100) + 2 \times (1/1000)$.</i> b. Compare two decimals to thousandths based on meanings of the digits in each place, using $>$, $=$, and $<$ symbols to record the results of comparisons.	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F
5.NBT.4	Use place value understanding to round decimals to any place.	Mission 1, Topic C
5.NBT.5	Fluently multiply multi-digit whole numbers using the standard algorithm.	Mission 2, Topic B Mission 2, Topic D
5.NBT.6	Find whole-number quotients of whole numbers with up to four-digit dividends and two-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 2, Topic E Mission 2, Topic F Mission 2, Topic H
5.NBT.7	Add, subtract, multiply, and divide decimals to hundredths, using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction; relate the strategy to a written method and explain the reasoning used. In this standard, dividing decimals is limited to a whole number dividend with a decimal divisor or a decimal dividend with a whole number divisor. Compare the value of the quotient on the basis of the values of the dividend and divisor.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 2, Topic C Mission 2, Topic D Mission 2, Topic G Mission 2, Topic H Mission 4, Topic E Mission 4, Topic G
Numbers & Operations - Fractions		

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
5.NF.1	Add and subtract fractions with unlike denominators (including mixed numbers) by replacing given fractions with equivalent fractions in such a way as to produce an equivalent sum or difference of fractions with like denominators. <i>For example, $\frac{2}{3} + \frac{5}{4} = \frac{8}{12} + \frac{15}{12} = \frac{23}{12}$. (In general, $\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$.)</i>	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D
5.NF.2	Solve real-world problems involving addition and subtraction of fractions referring to the same whole, including cases of unlike denominators by, <i>for example, using visual fraction models or equations to represent the problem</i> . Use benchmark fractions and number sense of fractions to estimate mentally and assess the reasonableness of answers. <i>For example, recognize $\frac{2}{5} + \frac{1}{2} = \frac{3}{7}$ as an incorrect result, by observing that $\frac{3}{7} < \frac{1}{2}$.</i>	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 4, Topic D
5.NF.3	Interpret a fraction as division of the numerator by the denominator ($\frac{a}{b} = a \div b$). Solve real-world problems involving division of whole numbers leading to answers in the form of fractions or mixed numbers, through the use of visual fraction models or equations to represent the problem. <i>For example, interpret $\frac{3}{4}$ as the result of dividing three by four, noting that $\frac{3}{4}$ multiplied by four equals three, and that when three wholes are shared equally among four people each person has a share of size $\frac{3}{4}$. If nine people want to share a 50-pound sack of rice equally by weight, how many pounds of rice should each person get? Between what two whole numbers does your answer lie?</i>	Mission 3, Topic A Mission 4, Topic B
5.NF.4	Apply and extend previous understandings of multiplication to multiply a fraction or whole number by a fraction. a. Interpret the product $(\frac{a}{b}) \times q$ as a parts of a partition of q into b equal parts; equivalently, as the result of a sequence of operations $a \times q \div b$ using a visual fraction model. <i>For example, use a fraction model to show $(\frac{2}{3}) \times 4 = \frac{8}{3}$, and create a story context for this equation. Do the same with $(\frac{2}{3}) \times (\frac{4}{5}) = \frac{8}{15}$. (In general, $(\frac{a}{b}) \times (\frac{c}{d}) = \frac{ac}{bd}$.)</i> b. Find the area of a rectangle with fractional side lengths by tiling it with unit squares of the appropriate unit fraction side lengths, and show that the area is the same as would be found by multiplying the side	Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 5, Topic C

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
	lengths. Multiply fractional side lengths to find areas of rectangles, and represent fraction products as rectangular areas.	
5.NF.5	Interpret multiplication as scaling. a. Compare the size of a product to the size of one factor on the basis of the size of the other factor, without performing the indicated multiplication. <i>For example, the products of expressions such as 5×3 or $\frac{1}{2} \times 3$ can be interpreted in terms of a quantity, three, and a scaling factor, five or $\frac{1}{2}$. Thus in addition to knowing that $5 \times 3 = 15$, they can also say that 5×3 is five times as big as three, without evaluating the product. Likewise they see $\frac{1}{2} \times 3$ as half the size of three.</i> b. Explain why multiplying a given number by a fraction greater than one results in a product greater than the given number (recognizing multiplication by whole numbers greater than one as a familiar case); explain why multiplying a given number by a fraction less than one results in a product smaller than the given number; and relate the principle of fraction equivalence. <i>For example, $\frac{6}{10} = \frac{(2 \times 3)}{(2 \times 5)}$. In general, $\frac{a}{b} = \frac{(n \times a)}{(n \times b)}$ has the effect of multiplying $\frac{a}{b}$ by one.</i>	Mission 4, Topic F
5.NF.6	Solve real-world problems involving multiplication of fractions and mixed numbers, <i>for example, by using visual fraction models or equations to represent the problem.</i>	Mission 4, Topic D Mission 4, Topic E Mission 4, Topic F Mission 5, Topic C
5.NF.7	Apply and extend previous understandings of division to divide unit fractions by whole numbers and whole numbers by unit fractions. Use strategies to divide fractions by reasoning about the relationship between multiplication and division. Division of a fraction by a fraction is not a requirement at this grade. a. Interpret division of a unit fraction by a non-zero whole number, and compute such quotients. <i>For example, create a story context for $(\frac{1}{3}) \div 4$, and use a visual fraction model to show the quotient. Use the relationship between multiplication and division to explain that $(\frac{1}{3}) \div 4 = \frac{1}{12}$ because $(\frac{1}{12}) \times 4 = \frac{1}{3}$.</i> b. Interpret division of a whole number by a unit fraction, and compute such quotients. <i>For example, create a story context for $4 \div (\frac{1}{5})$, and use a visual fraction model to</i>	Mission 4, Topic G

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
show the quotient. Use the relationship between multiplication and division to explain that $4 \div (1/5) = 20$ because $20 \times (1/5) = 4$. c. Solve real-world problems involving division of unit fractions by non-zero whole numbers and division of whole numbers by unit fractions, for example, by using visual fraction models and equations to represent the problem. For example, how much chocolate will each person get if three people share $1/2$ lb. of chocolate equally? How many $1/3$-cup servings are in two cups of raisins?		
Measurement & Data		
5.MD.1	Convert among different-sized standard measurement units within a given measurement system (for example, convert 5 cm to 0.05 m); use these conversions in solving multi-step, real-world problems.	Mission 1, Topic A Mission 2, Topic D Mission 4, Topic C Mission 4, Topic E
5.MD.2	Make a line plot to display a data set of measurements in fractions of a unit (halves, quarters, eighths). Use operations on fractions for this grade to solve problems involving information presented in line plots. For example, given graduated cylinders with different measures of liquid in each, find the amount of liquid each cylinder would contain if the total amount in all the cylinders were redistributed equally.	Mission 4, Topic A
5.MD.3	Recognize volume as an attribute of solid figures and understand concepts of volume measurement. a. A cube with side length one unit, called a "unit cube," is said to have "one cubic unit" of volume, and can be used to measure volume. b. A solid figure which can be packed without gaps or overlaps using n unit cubes is said to have a volume of n cubic units.	Mission 5, Topic A Mission 5, Topic B
5.MD.4	Measure volumes by counting unit cubes, using cubic cm, cubic in., cubic ft., and improvised units.	Mission 5, Topic A
5.MD.5	Relate volume to the operations of multiplication and addition and solve real-world and mathematical problems involving volume. a. Find the volume of a right rectangular prism with whole-number side lengths by packing it with unit cubes, and show that the volume is the same as would be found by	Mission 5, Topic B

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
multiplying the edge lengths, equivalently by multiplying the height by the area of the base. Represent threefold whole-number products as volumes, <i>for example, to represent the associative property of multiplication</i>. b. Apply the formulas $V = l \times w \times h$ and $V = b \times h$ for rectangular prisms to find volumes of right rectangular prisms with whole-number edge lengths in the context of solving real-world and mathematical problems. c. Recognize volume as additive. Find volumes of solid figures composed of two nonoverlapping right rectangular prisms by adding the volumes of the non-overlapping parts, applying this technique to solve real-world problems.		
Geometry		
5.G.1	Compose and understand the coordinate plane. a. Use a pair of perpendicular number lines, called axes, to define a coordinate system, with the intersection of the lines (the origin) arranged to coincide with the zero on each line and a given point in the plane located by using an ordered pair of numbers, called its coordinates. b. Using quadrant one on the coordinate plane, understand that the first number in a coordinate pair indicates how far to travel from the origin in the direction of the horizontal axis, and the second number indicates how far to travel in the direction of the vertical axis, with the convention that the names of the two axes and the coordinates correspond (x-axis and x-coordinate, y-axis and y-coordinate).	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C
5.G.2	Represent real-world and mathematical problems by graphing points in the first quadrant of the coordinate plane, and interpret coordinate values of points in the context of the situation.	Mission 6, Topic C Mission 6, Topic D
5.G.3	Understand that attributes belonging to a category of two-dimensional figures also belong to all subcategories of that category. For example, all rectangles have four right angles and all squares are rectangles, so all squares have four right angles.	Mission 5, Topic D

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
5.G.4	Classify two-dimensional figures in a hierarchy based on properties.	Mission 5, Topic D

6th Grade

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
Ratios & Proportional Relationships		
6.RP.1	Understand the concept of a ratio and use ratio language to describe a ratio relationship between two quantities. The following are examples of ratio language: <i>"The ratio of wings to beaks in the bird house at the zoo was 2:1, because for every two wings there was one beak."</i> <i>"For every vote candidate A received, candidate C received nearly three votes."</i>	Mission 2, Topic A Mission 2, Topic B
6.RP.2	Understand the concept of a unit rate a/b associated with a ratio $a:b$ with $b \neq 0$, and use rate language in the context of a ratio relationship. The following are examples of rate language: <i>"This recipe has a ratio of four cups of flour to two cups of sugar, so the rate is two cups of flour for each cup of sugar."</i> <i>"We paid \$75 for 15 hamburgers, which is a rate of \$5 per hamburger."</i> (In sixth grade, unit rates are limited to non-complex fractions.)	Mission 2, Topic C Mission 3, Topic A Mission 3, Topic C
6.RP.3	Use ratio and rate reasoning to solve real-world (with a context) and mathematical (void of context) problems, using strategies such as reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations involving unit rate problems. a. Make tables of equivalent ratios relating quantities with whole-number measurements, find missing values in the tables, and plot the pairs of values on the coordinate plane. Use tables to compare ratios. b. Solve unit rate problems including those involving unit pricing and constant speed. <i>For example, if it took four hours to mow eight lawns, how many lawns could be mowed in 32 hours? What is the hourly rate at which lawns were being mowed?</i> c. Find a percent of a quantity as a rate per 100. Solve problems involving finding the whole, given a part and the percent. <i>(For example, 30% of a quantity means 30/100 times the quantity.)</i> d. Use ratio reasoning to convert measurement units; manipulate and transform units appropriately when multiplying or dividing quantities.	Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E Mission 2, Topic F Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 6, Topic B Mission 6, Topic D

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
The Number System		
6.NS.1	Interpret and compute quotients of fractions. a. Compute quotients of fractions by fractions, <i>for example, by applying strategies such as visual fraction models, equations, and the relationship between multiplication and division, to represent problems.</i> b. Solve real-world problems involving division of fractions by fractions. <i>For example, how much chocolate will each person get if three people share $\frac{1}{2}$ pound of chocolate equally? How many $\frac{3}{4}$-cup servings are in $\frac{2}{3}$ of a cup of yogurt? How wide is a rectangular strip of land with length $\frac{3}{4}$ mile and area $\frac{1}{2}$ square mile?</i> c. Explain the meaning of quotients in fraction division problems. <i>For example, create a story context for $(\frac{2}{3}) \div (\frac{3}{4})$ and use a visual fraction model to show the quotient. Use the relationship between multiplication and division to explain that $(\frac{2}{3}) \div (\frac{3}{4}) = \frac{8}{9}$ because $\frac{3}{4}$ of $\frac{8}{9}$ is $\frac{2}{3}$. (In general, $(\frac{a}{b}) \div (\frac{c}{d}) = \frac{ad}{bc}$.)</i>	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E
6.NS.2	Fluently divide multi-digit numbers using the standard algorithm. a. Fluently divide multi-digit decimals using the standard algorithm, limited to a whole number dividend with a decimal divisor or a decimal dividend with a whole number divisor. b. Solve division problems in which both the dividend and the divisor are multi-digit decimals; develop the standard algorithm by using models, the meaning of division, and place value understanding.	Mission 5, Topic D
6.NS.3	Fluently add, subtract, multiply, and divide multi-digit decimals using the standard algorithm for each operation.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 6, Topic A Mission 8, Topic C
6.NS.4	Find the greatest common factor of two whole numbers less than or equal to 100 and the least common multiple of two whole numbers less than or equal to 12. Use the distributive property to express a sum of two whole numbers 1–100 with a common factor as a multiple of a sum of two whole numbers	Mission 7, Topic D

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	with no common factor. <i>For example, express $36 + 8$ as $4(9 + 2)$.</i>	
6.NS.5	Understand that positive and negative numbers are used together to describe quantities having opposite directions or values (<i>for example, temperature above/below zero, elevation above/below sea level, credits/debits, positive/negative electric charge</i>); use positive and negative numbers to represent quantities in real-world contexts, explaining the meaning of zero in each situation.	Mission 7, Topic A
6.NS.6	Understand a rational number as a point on the number line. Extend number line diagrams and coordinate axes familiar from previous grades to represent points on the line and in the plane with negative number coordinates. a. Recognize opposite signs of numbers as indicating locations on opposite sides of zero on the number line; recognize that the opposite of the opposite of a number is the number itself. <i>For example, $-(-3) = 3$, and zero is its own opposite.</i> b. Understand that the signs of numbers in ordered pairs indicate their location in quadrants of the coordinate plane; recognize that when two ordered pairs differ only by signs, the locations of the points are related by reflections across one or both axes. c. Find and position integers and other rational numbers on a horizontal or vertical number line diagram; find and position pairs of integers and other rational numbers on a coordinate plane.	Mission 7, Topic A Mission 7, Topic C
6.NS.7	Understand ordering and absolute value of rational numbers. a. Interpret statements of inequality as statements about the relative position of two numbers on a number line diagram. <i>For example, interpret $-3 > -7$ as a statement that -3 is located to the right of -7 on a number line oriented from left to right.</i> b. Write, interpret, and explain statements of order for rational numbers in real-world contexts. <i>For example, write $-3^{\circ}\text{C} > -7^{\circ}\text{C}$ to express the fact that -3°C is warmer than -7°C.</i> c. Understand the absolute value of a rational number as its distance from zero on the number line; interpret absolute value as magnitude for a positive or negative quantity in a real-world context. <i>For example,</i>	Mission 7, Topic A Mission 7, Topic B Mission 7, Topic C

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	for an account balance of -30 dollars, write $ -30 = 30$ to describe the size of the debt in dollars. d. Distinguish comparisons of absolute value from statements about order. For example: Recognize that an account balance less than -30 dollars represents a debt greater than 30 dollars.	
6.NS.8	Solve real-world and mathematical problems by graphing points in all four quadrants of the coordinate plane. Include use of coordinates and absolute value to find distances between points with the same x -coordinate or the same y -coordinate.	Mission 7, Topic C
Expressions & Equations		
6.EE.1	Write and evaluate numerical expressions involving whole-number exponents.	Mission 1, Topic F Mission 6, Topic C
6.EE.2	Write, read, and evaluate expressions in which letters represent numbers. a. Write expressions that record operations with numbers and with letters representing numbers. For example, express the calculation "Subtract y from 5" as $5 - y$ and express "Jane had \$105.00 in her bank account. One year later, she had x dollars more. Write an expression that shows her new balance" as $\$105.00 + x$. b. Identify parts of an expression using mathematical terms (for example, sum, term, product, factor, quotient, coefficient); view one or more parts of an expression as a single entity and a sum of two terms. For example, describe the expression $2(8 + 7)$ as a product of two factors; view $(8 + 7)$ as both a single entity and a sum of two terms. c. Evaluate expressions at specific values of their variables. Include expressions that arise from formulas used in real-world problems. Perform arithmetic operations, including those involving whole-number exponents, applying the Order of Operations when there are no parentheses to specify a particular order. For example, use the formulas $V = s^3$ and $A = 6s^2$ to find the volume and surface area of a cube with sides of length $s = 1/2$.	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic F Mission 6, Topic B Mission 6, Topic C Mission 7, Topic B
6.EE.3	Apply the properties of operations to generate equivalent expressions. For example, apply the	Mission 6, Topic B

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	<i>distributive property to the expression $3(2 + x)$ to produce the equivalent expression $6 + 3x$; apply the distributive property to the expression $24x + 18y$ to produce the equivalent expression $6(4x + 3y)$; apply properties of operations to $y + y + y$ to produce the equivalent expression $3y$.</i>	
6.EE.4	Identify when two expressions are equivalent. <i>For example, the expressions $y + y + y$ and $3y$ are equivalent because they name the same number, regardless of which number y represents.</i>	Mission 5, Topic D Mission 6, Topic B
6.EE.5	Understand solving an equation or inequality as a process of answering the question: Which values from a specified set, if any, make the equation or inequality true? Use substitution to determine whether a given number in a specified set makes an equation or inequality true.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 7, Topic B
6.EE.6	Use variables to represent numbers and write expressions when solving a real-world or mathematical problem; understand that a variable can represent an unknown number, or, depending on the purpose at hand, any number in a specified set.	Mission 6, Topic A Mission 6, Topic B Mission 7, Topic B
6.EE.7	Solve real-world and mathematical problems by writing and solving equations of the form $x + a = b$ and $ax = b$ for cases in which a , b and x are all non-negative rational numbers.	Mission 6, Topic A Mission 6, Topic B
6.EE.8	Write an inequality of the form $x > c$ or $x < c$ to represent a constraint or condition in a real-world or mathematical problem. Recognize that inequalities of the form $x > c$ or $x < c$ have infinitely many solutions; represent solutions of such inequalities on number line diagrams.	Mission 7, Topic B
6.EE.9	Use variables to represent two quantities in a real-world problem that change in relationship to one another; write an equation to express one quantity, thought of as the dependent variable, in terms of the other quantity, thought of as the independent variable. Analyze the relationship between the dependent and independent variables using graphs	Mission 6, Topic D

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
and tables, and relate these to the equation. <i>For example, in a problem involving motion at constant speed, list and graph ordered pairs of distances and times, and write the equation $d = 65t$ to represent the relationship between distance and time</i>		
Geometry		
6.G.1	Find the area of right triangles, other triangles, special quadrilaterals, and polygons by composing and decomposing into rectangles, triangles and/or other shapes; apply these techniques in the context of solving real-world and mathematical problems.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic G Mission 4, Topic D
6.G.2	Find the volume of a right rectangular prism with appropriate unit fraction edge lengths by packing it with cubes of the appropriate unit fraction edge lengths (<i>for example, $3\frac{1}{2} \times 2 \times 6$</i>), and show that the volume is the same as would be found by multiplying the edge lengths of the prism. Apply the formulas $V = lwh$ and $V = bh$ to find volumes of right rectangular prisms with fractional edge lengths in the context of solving real-world and mathematical problems. (Note: Model the packing using drawings and diagrams.)	Mission 1, Topic E Mission 4, Topic D Mission 4, Topic E
6.G.3	Draw polygons in the coordinate plane given coordinates for the vertices; use coordinates to find the length of a side joining points with the same x coordinate or the same y coordinate. Apply these techniques in the context of solving real-world and mathematical problems.	Mission 7, Topic C
6.G.4	Represent three-dimensional figures using nets made up of rectangles and triangles, and use the nets to find the surface area of these figures. Apply these techniques in the context of solving real-world and mathematical problems.	Mission 1, Topic E Mission 1, Topic F Mission 1, Topic G
Statistics & Probability		
6.SP.1	Recognize a statistical question as one that anticipates variability in the data related to the question and	Mission 8, Topic A Mission 8, Topic B

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	accounts for it in the answers. <i>For example, "How old am I?" is not a statistical question, but "How old are the students in my school?" is a statistical question because one anticipates variability in students' ages.</i>	Mission 8, Topic D
6.SP.2	Understand that a set of data collected to answer a statistical question has a distribution that can be described by its center, spread/range and overall shape.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic E
6.SP.3	Recognize that a measure of center for a numerical data set summarizes all of its values with a single number, while a measure of variation describes how its values vary with a single number.	Mission 8, Topic B
6.SP.4	Display numerical data in plots on a number line, including dot plots, histograms and box plots. Choose the most appropriate graph/plot for the data collected.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D
6.SP.5	Summarize numerical data sets in relation to their context, such as by: a. Reporting the number of observations. b. Describing the nature of the attribute under investigation, including how it was measured and its units of measurement. c. Giving quantitative measures of center (median and/or mean) and variability (interquartile range and/or mean absolute deviation), as well as describing any overall pattern and any striking deviations (for example, outliers) from the overall pattern with reference to the context in which the data were gathered. d. Relating the choice of measures of center and variability to the shape of the data distribution and the context in which the data were gathered.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E

7th Grade

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
Ratios & Proportional Relationships		
7.RP.1	Compute unit rates associated with ratios of fractions, including ratios of lengths, areas and other quantities measured in like or different units. <i>For example, if a person walks $\frac{1}{2}$ mile in each $\frac{1}{4}$ hour, compute the unit rate as the complex fraction $\frac{1/2}{1/4}$ miles per hour, equivalently 2 miles per hour.</i>	Mission 2, Topic C Mission 4, Topic A
7.RP.2	Recognize and represent proportional relationships between quantities. a. Decide whether two quantities are in a proportional relationship, e.g., by testing for equivalent ratios in a table or graphing on a coordinate plane and observing whether the graph is a straight line through the origin. b. Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships. c. Represent proportional relationships by equations. <i>For example, if total cost t is proportional to the number n of items purchased at a constant price p, the relationship between the total cost and the number of items can be expressed as $t = pn$.</i> d. Explain what a point (x, y) on the graph of a proportional relationship means in terms of the situation, with special attention to the points $(0, 0)$ and $(1, r)$ where r is the unit rate.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E Mission 3, Topic A Mission 3, Topic B Mission 4, Topic A Mission 5, Topic C Mission 5, Topic D
7.RP.3	Use proportional relationships to solve multi-step ratio and percent problems. <i>Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.</i>	Mission 3, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
The Number System		
7.NS.1	Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers; represent addition and subtraction on a horizontal or vertical number line diagram. a. Describe situations in which opposite quantities combine to make 0. <i>For</i>	Mission 5, Topic A Mission 5, Topic B Mission 6, Topic D Mission 7, Topic B

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
	<i>example, a hydrogen atom has 0 charge because its two constituents are oppositely charged.</i> b. Understand $p + q$ as the number located a distance q from p in the positive or negative direction, depending on whether q is positive or negative. Show that a number and its opposite have a sum of 0 (are additive inverses). Interpret sums of rational numbers by describing real-world contexts. c. Understand subtraction of rational numbers as adding the additive inverse, $p - q = p + (-q)$. Show that the distance between two rational numbers on the number line is the absolute value of their difference, and apply this principle in real-world contexts. d. Apply properties of operations as strategies to add and subtract rational numbers.	
7.NS.2	Apply and extend previous understandings of multiplication and division and of fractions to multiply and divide rational numbers. a. Understand that multiplication is extended from fractions to rational numbers by requiring that operations continue to satisfy the properties of operations, particularly the distributive property, leading to products such as $(-1)(-1) = 1$ and the rules for multiplying signed numbers. Interpret products of rational numbers by describing real-world contexts. b. Understand that integers can be divided, provided the divisor is not zero, and that every quotient of integers (with non-zero divisor) is a rational number. If p and q are integers, then $-(p/q) = (-p)/q = p/(-q)$. Interpret quotients of rational numbers by describing real-world contexts. c. Apply properties of operations as strategies to multiply and divide rational numbers. d. Convert a rational number to a decimal using long division; know that the decimal form of a rational number terminates in 0s or eventually repeats.	Mission 4, Topic A Mission 5, Topic A Mission 5, Topic C Mission 8, Topic D
7.NS.3	Solve real-world and mathematical problems involving the four operations with rational numbers. Computations with rational numbers extend the rules for manipulating fractions to complex fractions.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
Expression & Equations		

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.EE.1	Apply properties of operations as strategies to add, subtract, factor, and expand linear expressions with rational coefficients.	Mission 6, Topic D
7.EE.2	Understand that rewriting an expression in different forms in a problem context can shed light on the problem, and how the quantities in it are related. <i>For example, $a + 0.05a = 1.05a$ means that “increase by 5%” is the same as “multiply by 1.05.”</i>	Mission 6, Topic B
7.EE.3	Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form, convert between forms as appropriate, and assess the reasonableness of answers using mental computation and estimation strategies. <i>For example: If a woman making \$25 an hour gets a 10% raise, she will make an additional $\frac{1}{10}$ of her salary an hour, or \$2.50, for a new salary of \$27.50. If you want to place a towel bar $9\frac{3}{4}$ inches long in the center of a door that is $27\frac{1}{2}$ inches wide, you will need to place the bar about 9 inches from each edge; this estimate can be used as a check on the exact computation.</i>	Mission 3, Topic C Mission 5, Topic C Mission 5, Topic F Mission 6, Topic A Mission 6, Topic B
7.EE.4	Use variables to represent quantities in a real-world or mathematical problem, and construct simple equations and inequalities to solve problems by reasoning about the quantities. a. Solve word problems leading to equations of the form $px + q = r$ and $p(x + q) = r$, where p , q , and r are specific rational numbers. Solve equations of these forms fluently. Compare an algebraic solution to an arithmetic solution, identifying the sequence of the operations used in each approach. <i>For example, the perimeter of a rectangle is 54 cm. Its length is 6 cm. What is its width?</i> b. Solve word problems leading to inequalities of the form $px + q > r$ or $px + q < r$, where p , q , and r are specific rational numbers. Graph the solution set of the inequality and interpret it in the context of the problem. <i>For example: As a salesperson, you are paid \$50 per week plus \$3 per sale. This week you want your pay to be at least \$100. Write an inequality for the</i>	Mission 5, Topic E Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 7, Topic A

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
<i>number of sales you need to make, and describe the solutions.</i>		
Geometry		
7.G.1	Solve problems involving scale drawings of geometric figures, such as computing actual lengths and areas from a scale drawing and reproducing a scale drawing at a different scale.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 2, Topic A Mission 3, Topic B Mission 3, Topic C
7.G.2	Draw (freehand, with ruler and protractor, and with technology) geometric shapes with given conditions. Focus on constructing triangles from three measures of angles or sides, noticing when the conditions determine a unique triangle, more than one triangle, or no triangle.	Mission 3, Topic A Mission 7, Topic B Mission 7, Topic D
7.G.3	Describe the two-dimensional figures that result from slicing three-dimensional figures, as in plane sections of right rectangular prisms and right rectangular pyramids.	Mission 7, Topic C
7.G.4	Know the formulas for the area and circumference of a circle, and solve problems; give an informal derivation of the relationship between the circumference and area of a circle.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C
7.G.5	Use facts about supplementary, complementary, vertical, and adjacent angles in a multi-step problem to write, and use them to solve simple equations for an unknown angle in a figure.	Mission 7, Topic A
7.G.6	Solve real-world and mathematical problems involving area, volume and surface area of two- and three-dimensional objects composed of triangles, quadrilaterals, polygons, cubes, and right prisms.	Mission 1, Topic A Mission 2, Topic C Mission 3, Topic B Mission 7, Topic C Mission 7, Topic D
Statistics & Probability		

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.SP.1	Understand that statistics can be used to gain information about a population by examining a sample of the population, and that generalizations about a population from a sample are valid only if the sample is representative of that population. Understand that random sampling is more likely to produce representative samples and support valid inferences.	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
7.SP.2	Use data from a random sample to draw inferences about a population with an unknown characteristic of interest. Generate multiple samples (or simulated samples) of the same size to gauge the variation in estimates or predictions. <i>For example, estimate the mean word length in a book by randomly sampling words from the book; predict the winner of a school election based on randomly sampled survey data. Gauge how far off the estimate or prediction might be.</i>	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
7.SP.3	Informally assess the degree of visual overlap of two numerical data distributions with similar variabilities, estimating the difference between the centers by expressing it as a multiple of a measure of variability. <i>For example, the mean height of players on the basketball team is 10 cm greater than the mean height of players on the soccer team, approximately twice the variability (mean absolute deviation) on either team; on a dot plot, the separation between the two distributions of heights is noticeable.</i>	Mission 8, Topic C Mission 8, Topic D
7.SP.4	Use measures of center and measures of variability for numerical data from random samples to draw informal comparative inferences about two populations. <i>For example, decide whether the words in a chapter of a seventh grade science book are generally longer than the words in a chapter of a fourth grade science book.</i>	Mission 8, Topic D Mission 8, Topic E
7.SP.5	Understand that the probability of a chance event is a number between 0 and 1 that expresses the likelihood of the event occurring. Larger numbers indicate greater likelihood. A probability near 0 indicates an unlikely event, a probability around $\frac{1}{2}$ indicates an	Mission 8, Topic A

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
	event that is neither unlikely nor likely, and a probability near 1 indicates a likely event.	
7.SP.6	Approximate the probability of a chance event by collecting data on the chance process that produces it and observing its long-run relative frequency, and predict the approximate relative frequency given the probability. <i>For example, when rolling a number cube 600 times, predict that a 3 or 6 would be rolled roughly 200 times, but probably not exactly 200 times.</i>	Mission 8, Topic A
7.SP.7	Develop a probability model and use it to find probabilities of events. Compare probabilities from a model to observed frequencies; if the agreement is not good, explain possible sources of the discrepancy. a. Develop a uniform probability model by assigning equal probability to all outcomes, and use the model to determine probabilities of events. <i>For example, if a student is selected at random from a class, find the probability that Jane will be selected and the probability that a girl will be selected.</i> b. Develop a probability model (which may not be uniform) by observing frequencies in data generated from a chance process. <i>For example, find the approximate probability that a spinning penny will land heads up or that a tossed paper cup will land open-end down. Do the outcomes for the spinning penny appear to be equally likely based on the observed frequencies?</i>	Mission 8, Topic A Mission 8, Topic C Mission 8, Topic E
7.SP.8	Find probabilities of compound events using organized lists, tables, tree diagrams, and simulation. a. Understand that, just as with simple events, the probability of a compound event is the fraction of outcomes in the sample space for which the compound event occurs. b. Represent sample spaces for compound events using methods such as organized lists, tables and tree diagrams. For an event described in everyday language (e.g., “rolling double sixes”), identify the outcomes in the sample space which compose the event. c. Design and use a simulation to generate frequencies for compound events. <i>For example, use random digits as a simulation tool to approximate the answer to the question: If 40% of donors have type A blood, what is the probability</i>	Mission 8, Topic A Mission 8, Topic B

UTAH CORE STATE STANDARDS FOR MATHEMATICS	ZEARN MATH
7th Grade Standards	Lessons
<i>that it will take at least 4 donors to find one with type A blood?</i>	

8th Grade

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
The Number System		
8.NS.1	Know that numbers that are not rational are called irrational. Understand informally that every number has a decimal expansion; for rational numbers, show that the decimal expansion repeats eventually, and convert a decimal expansion which repeats eventually into a rational number.	Mission 8, Topic E
8.NS.2	Use rational approximations of irrational numbers to compare the size of irrational numbers, locate them approximately on a number line diagram, and estimate the value of expressions (e.g., π^2). <i>For example, by truncating the decimal expansion of $\sqrt{2}$, show that $\sqrt{2}$ is between 1 and 2, then between 1.4 and 1.5, and explain how to continue on to get better approximations.</i>	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
8.NS.3	Understand how to perform operations and simplify radicals with emphasis on square roots.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Expressions & Equations		
8.EE.1	Know and apply the properties of integer exponents to generate equivalent numerical expressions. <i>For example, $32 \times 3^{-5} = 3^{-3} = 1/33 = 1/27$.</i>	Mission 7, Topic B Mission 7, Topic C
8.EE.2	Use square root and cube root symbols to represent solutions to equations of the form $x^2 = p$ and $x^3 = p$, where p is a positive rational number. Evaluate square roots of small perfect squares and cube roots of small perfect cubes. Know that $\sqrt{2}$ is irrational.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D
8.EE.3	Use numbers expressed in the form of a single digit times an integer power of 10 to estimate very large or very small quantities, and to express how many times as much one is than the other. <i>For example, estimate the population of the United States as 3×10^8 and the</i>	Mission 7, Topic C Mission 7, Topic D

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
	<i>population of the world as 7×10^9, and determine that the world population is more than 20 times larger.</i>	
8.EE.4	Perform operations with numbers expressed in scientific notation, including problems where both decimal and scientific notation are used. Use scientific notation and choose units of appropriate size for measurements of very large or very small quantities (e.g., use millimeters per year for seafloor spreading). Interpret scientific notation that has been generated by technology.	Mission 7, Topic C Mission 7, Topic D
8.EE.5	Graph proportional relationships, interpreting the unit rate as the slope of the graph. Compare two different proportional relationships represented in different ways. <i>For example, compare a distance-time graph to a distance-time equation to determine which of two moving objects has greater speed.</i>	Mission 3, Topic A Mission 3, Topic B
8.EE.6	Use similar triangles to explain why the slope m is the same between any two distinct points on a non-vertical line in the coordinate plane; derive the equation $y = mx$ for a line through the origin and the equation $y = mx + b$ for a line intercepting the vertical axis at b .	Mission 2, Topic C Mission 3, Topic B Mission 3, Topic C Mission 3, Topic E
8.EE.7	Solve linear equations and inequalities in one variable. a. Give examples of linear equations in one variable with one solution, infinitely many solutions, or no solutions. Show which of these possibilities is the case by successively transforming the given equation into simpler forms, until an equivalent equation of the form $x = a$, $a = a$, or $a = b$ results (where a and b are different numbers). b. Solve single-variable linear equations and inequalities with rational number coefficients, including equations and inequalities whose solutions require expanding expressions using the distributive property and collecting like terms. c. Solve single-variable absolute value equations.	Mission 4, Topic B Grade 7, Mission 6, Topic C
8.EE.8	Analyze and solve pairs of simultaneous linear equations. a. Understand that solutions to a system of two linear equations in two variables correspond to	Mission 3, Topic D Mission 3, Topic E Mission 4, Topic B

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
	points of intersection of their graphs, because points of intersection satisfy both equations simultaneously. b. Solve systems of two linear equations in two variables graphically, approximating when solutions are not integers and estimate solutions by graphing the equations. Solve simple cases by inspection. <i>For example, $3x + 2y = 5$ and $3x + 2y = 6$ have no solution because $3x + 2y$ cannot simultaneously be 5 and 6.</i> c. Solve real-world and mathematical problems leading to two linear equations in two variables graphically. <i>For example, given coordinates for two pairs of points, determine whether the line through the first pair of points intersects the line through the second pair.</i>	Mission 4, Topic C Mission 4, Topic D
Functions		
8.F.1	Understand that a function is a rule that assigns to each input exactly one output. The graph of a function is the set of ordered pairs consisting of an input and the corresponding output. (Function notation is not required in grade 8.)	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic E
8.F.2	Compare properties of two functions, each represented in a different way (algebraically, graphically, numerically in tables, or by verbal descriptions). <i>For example, given a linear function represented by a table of values and a linear function represented by an algebraic expression, determine which function has the greater rate of change.</i>	Mission 5, Topic B Mission 5, Topic C
8.F.3	Interpret the equation $y = mx + b$ as defining a linear function whose graph is a straight line; give examples of functions that are not linear. <i>For example, the function $A = s^2$, giving the area of a square as a function of its side length, is not linear because its graph contains the points (1,1), (2,4) and (3,9), which are not on a straight line.</i>	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic E
8.F.4	Construct a function to model a linear relationship between two quantities. Determine the rate of change and initial value of the function from a description of a relationship or from two (x, y) values, including reading these from a table or from a graph. Interpret the rate of change and initial value of a linear function	Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
	in terms of the situation it models, and in terms of its graph or a table of values.	
8.F.5	Describe qualitatively the functional relationship between two quantities by analyzing a graph (e.g., where the function is increasing or decreasing, linear or nonlinear). Sketch a graph that exhibits the qualitative features of a function that has been described verbally.	Mission 5, Topic B Mission 5, Topic C
Geometry		
8.G.1	Verify experimentally the properties of rotations, reflections, and translations: a. Lines are taken to lines, and line segments to line segments of the same length. b. Angles are taken to angles of the same measure. c. Parallel lines are taken to parallel lines.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 3, Topic B
8.G.2	Understand that a two-dimensional figure is congruent to another if the second can be obtained from the first by a sequence of rotations, reflections, and translations; given two congruent figures, describe a sequence that exhibits the congruence between them.	Mission 1, Topic C Mission 1, Topic D Mission 2, Topic B
8.G.3	Observe that orientation of the plane is preserved in rotations and translations, but not with reflections. Describe the effect of dilations, translations, rotations, and reflections on two-dimensional figures using coordinates.	Mission 1, Topic A Mission 2, Topic A Mission 2, Topic C
8.G.4	Understand that a two-dimensional figure is similar to another if the second can be obtained from the first by a sequence of rotations, reflections, translations, and dilations; given two similar two-dimensional figures, describe a sequence that exhibits the similarity between them.	Mission 2, Topic B
8.G.5	Use informal arguments to establish facts about the angle sum and exterior angle of triangles, about the angles created when parallel lines are cut by a transversal, and the angle-angle criterion for similarity of triangles. <i>For example, arrange three copies of the</i>	Mission 1, Topic D Mission 2, Topic B Mission 2, Topic D

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
	<i>same triangle so that the sum of the three angles appears to form a line, and give an argument in terms of transversals why this is so.</i>	
8.G.6	Explore and explain proofs of the Pythagorean Theorem and its converse.	Mission 8, Topic C
8.G.7	Apply the Pythagorean Theorem to determine unknown side lengths in right triangles in real-world and mathematical problems in two and three dimensions.	Mission 8, Topic C
8.G.8	Apply the Pythagorean Theorem to find the distance between two points in a coordinate system.	Mission 8, Topic C
8.G.9	Know the formulas for the volumes of cones, cylinders, and spheres, and use them to solve real-world and mathematical problems.	Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
Statistics & Probability		
8.SP.1	Construct and interpret scatter plots for bivariate measurement data to investigate patterns of association between two quantities. Describe patterns such as clustering, outliers, positive or negative association, linear association, and nonlinear association.	Mission 6, Topic A Mission 6, Topic B
8.SP.2	Know that straight lines are widely used to model relationships between two quantitative variables. For scatter plots that suggest a linear association, informally fit a straight line, and informally assess the model fit by judging the closeness of the data points to the line.	Mission 6, Topic B
8.SP.3	Use the equation of a linear model to solve problems in the context of bivariate measurement data, interpreting the slope and intercept. <i>For example, in a linear model for a biology experiment, interpret a slope of 1.5 cm/hr as meaning that an additional hour of sunlight each day is associated with an additional 1.5 cm in mature plant height.</i> (Calculating equations for a linear model is not expected in grade 8.)	Mission 6, Topic B

UTAH CORE STATE STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
8.SP.4	Understand that patterns of association can also be seen in bivariate categorical data by displaying frequencies and relative frequencies in a two-way table. Construct and interpret a two-way table summarizing data on two categorical variables collected from the same subjects. Use relative frequencies calculated for rows or columns to describe possible association between the two variables. <i>For example, collect data from students in your class on whether or not they have a curfew on school nights and whether or not they have assigned chores at home. Is there evidence that those who have a curfew also tend to have chores?</i>	Mission 6, Topic C